

Notes: Benabou and Laroque (QJE, 1992)

Using Privileged Information to Manipulate Markets: Insiders, Gurus, and Credibility

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1 Big picture

- **Question.** Can people with privileged information manipulate asset markets by strategically distorting public announcements? If so, does reputation eventually discipline them, or can manipulation continue for a long time?
- **Core setting.** A privately informed agent, called the journalist, insider, manager, or guru, observes a noisy signal about an asset's payoff. He can send a public forecast that affects market beliefs and prices, and he can then trade on his private information.
- **Main mechanism.** Private information gives the sender both the ability and the incentive to manipulate. If his signal is good, he may announce bad news, depress the price, buy cheaply, and profit when the asset payoff is later revealed. If his signal is bad, he may announce good news, raise the price, sell or short, and profit when the truth comes out.
- **Why noisy information matters.** If insiders had perfect information, lies would be revealed immediately after the payoff realization. With noisy signals, a wrong forecast can be interpreted as an honest mistake. This makes manipulation harder to prove and allows reputations to decline gradually rather than collapse after one bad forecast.
- **Credibility as a strategic asset.** The public attaches credibility to the sender because he may be honest, and because even an opportunistic sender may sometimes tell the truth to preserve his reputation. Credibility is therefore a capital stock that can be invested in or spent down.
- **One-period result.** In a static game, messages influence prices only if the public's prior probability that the sender is honest is greater than one-half. If that prior is high enough, the opportunistic type always lies, but the message is still partly credible because it might have come from the honest type.
- **Repeated-game result.** In the infinite-horizon game, even very low-reputation senders retain some credibility as long as their reputation is not exactly zero. The opportunistic type always lies with positive probability, but his exact mixing depends on the reputational cost of being wrong.
- **Long-run learning.** If the sender's type is fixed forever, the public eventually learns it: an honest sender's reputation converges to one, and an opportunistic sender's reputation converges to zero. However, this learning can take a long time.
- **Persistent manipulation.** If types change over time, for example because different journalists or managers take over or because trading opportunities vary, reputation never fully reveals the current type. Manipulation remains possible indefinitely in the stationary long-run environment.
- **Policy message.** When truthfulness is difficult to verify, insider-trading restrictions or mandatory trade disclosure may be needed to preserve the integrity of information in prices. Reputation alone is not enough in the short and medium run, and may not be enough in the long run if types change.

2 Incremental contribution of the paper

- **From silent insider trading to active manipulation.** Much of the insider-trading literature studies informed traders who reveal information through trades. This paper studies informed

agents who deliberately affect public beliefs through announcements and then trade against those distorted beliefs.

- **Noisy private information and credibility.** The paper extends strategic-communication reputation models, especially Sobel’s credibility model, to the case of noisy private signals. This is crucial because noisy signals allow wrong forecasts to be explained as honest errors.
- **Repeated manipulation without immediate unraveling.** In perfect-information reputation games, one lie can permanently destroy credibility. Here, because the insider’s signal is imperfect, reputation moves up and down in a realistic way and manipulation can continue for many periods.
- **Credibility as endogenous market power.** The sender’s ability to move prices is not exogenously fixed. It depends on the public’s evolving belief that he is honest. The model links price responsiveness to reputation.
- **Two manipulation schemes.** The paper distinguishes pre-announcement speculation, where the sender trades before moving the price, from post-announcement speculation, where he first distorts beliefs and then trades on his private information. The main model focuses on the second, more information-distorting channel.
- **A clear policy rationale.** The model gives a theoretical basis for restrictions on trading by people who make market-moving forecasts, such as managers, financial journalists, and investment gurus. Trade restrictions are useful because truthfulness is harder to verify than trading.
- **Dynamic learning result.** The paper separates two long-run cases: fixed type and changing type. With a fixed type, truth eventually wins; with changing types, manipulation persists forever. This distinction is central for real-world institutions where office holders, analysts, or incentives change over time.
- **Broader applicability.** Although framed as a financial-market model, the theoretical mechanism applies to any setting with noisy private information, strategic announcements, and reputation: policy forecasts, firm disclosures, oligopoly information sharing, and public expert advice.

3 Institutional motivation and examples

3.1 Who can manipulate?

The paper has in mind informed agents whose public statements affect asset prices. The examples include:

- **Financial journalists.** A columnist may publish market-moving information while secretly trading directly or through associates. The paper cites the Winans–*Wall Street Journal* case as an obvious example.
- **Investment gurus.** A forecaster or newsletter writer may affect prices because many investors follow his forecasts. The paper mentions Robert Prechter as an example of a forecaster whose statements could move the market.
- **Corporate executives.** Managers often own or trade shares of their own firms while making public statements to investors, analysts, and shareholders.

3.2 Why standard reputation arguments are incomplete

A simple argument against manipulation is that a manipulator will be discovered, lose credibility, and face sanctions. The paper's key response is that this argument is weak when private information is noisy. If the sender's prediction is wrong, outsiders cannot easily tell whether the sender lied or honestly received a bad signal. Manipulation therefore damages reputation only gradually and probabilistically.

This feature makes the model very different from a perfect-information setting. With perfect signals, an incorrect prediction is proof of dishonesty. With noisy signals, an incorrect prediction is evidence, but not proof. The public updates beliefs using Bayes' rule, but reputation remains strictly between zero and one for a long time.

3.3 Examples of post-announcement manipulation

The paper highlights two corporate examples.

- **Texas Gulf Sulphur.** Company insiders allegedly learned of major mineral deposits, tried to convince the public that evidence was inconclusive, and accumulated shares and options before the discovery became public.
- **Emerson Radio.** Executives allegedly knew orders were falling and sales were weakening, but continued to make optimistic statements while insiders and associates sold shares.

The common feature is not simply that insiders traded before price-relevant news became public. The stronger point is that they allegedly distorted the information environment itself: they misled, delayed, or selectively disclosed information in order to influence prices.

4 Short-run model: assets, signals, and messages

4.1 Asset payoff and private signal

There is a continuum of agents trading a risky asset. The asset pays one dollar if the state is good and zero if the state is bad. The state is denoted

$$\tilde{n} \in \{-1, 1\}.$$

The good state is $\tilde{n} = 1$. The informed sender observes a private signal

$$\tilde{s} \in \{-1, 1\},$$

which predicts the state with accuracy

$$\Pr(\tilde{n} = n \mid \tilde{s} = n) = p > \frac{1}{2}, \quad n \in \{-1, 1\}.$$

The signal is informative but imperfect. This imperfectness is the source of the paper's main reputation result.

4.2 Sender types

The sender can be one of two types.

- **Honest type.** The honest type always reports the signal truthfully: $\tilde{m} = \tilde{s}$.

- **Opportunistic type.** The opportunistic type maximizes expected trading gains. He may tell the truth or lie depending on the current credibility incentives.

Let ρ denote the public's probability that the sender is honest. In the one-period game ρ is a prior. In the dynamic game it is the sender's reputation.

The opportunistic type uses a symmetric mixed reporting strategy:

$$q = \Pr(\tilde{m} = \tilde{s} \mid \tilde{s} = s), \quad s \in \{-1, 1\}.$$

Thus q is the probability that the opportunistic type tells the truth. The honest type has $q = 1$ by definition.

4.3 Credibility of a message

Credibility is the probability that the public forecast is correct. Given reputation ρ and opportunistic truth-telling probability q , credibility is

$$\pi = \rho p + (1 - \rho)[pq + (1 - p)(1 - q)].$$

The first term is the probability that the sender is honest and the signal is correct. The second term is the probability that the sender is opportunistic and his message happens to be correct.

A message $\tilde{m} = 1$ induces posterior belief

$$\beta = \Pr(\tilde{n} = 1 \mid \tilde{m} = 1) = \pi,$$

whereas a message $\tilde{m} = -1$ induces posterior belief

$$\beta = \Pr(\tilde{n} = 1 \mid \tilde{m} = -1) = 1 - \pi.$$

Thus the message moves public beliefs away from one-half whenever $\pi > 1/2$. If $\pi = 1/2$, the message has no informational content and prices do not respond.

4.4 Noise traders and random price response

The public consists of rational speculators and noise traders. Rational speculators use Bayes' rule. Noise traders act according to a random belief $\hat{\beta}$, which can be interpreted as a fad, liquidity shock, or random information shock. This noise matters for two reasons:

- It prevents the sender from being uniquely identified as the only trader who wants to trade at the prevailing price.
- It prevents riskless pre-announcement manipulation because the sender cannot perfectly predict how prices will react to his message.

Let $P(\beta)$ be the market-clearing price when rational traders hold belief β . The paper proves that the market-clearing price is unique and is increasing in rational traders' optimism. If β is higher, the distribution of prices shifts upward in the sense of first-order stochastic dominance.

5 Trading gains and two manipulation schemes

5.1 Post-announcement speculation

The main model assumes that the sender trades after the public announcement. The reason is to focus on the channel where the sender first changes beliefs and then trades against the price distortion.

Suppose the sender observes a good signal $\tilde{s} = 1$. If he truthfully announces good news, the public becomes optimistic, the price rises, and his expected trading gains are moderate. If he lies and announces $\tilde{m} = -1$, the public becomes pessimistic, the price falls, and he can buy the underpriced asset. Since the asset is more likely to pay off when $\tilde{s} = 1$, this gives him a strong incentive to lie.

Similarly, if the signal is bad $\tilde{s} = -1$, the sender benefits from announcing good news, raising the price, and selling or shorting the asset.

5.2 Expected utility from a message

Let $U(s, \beta)$ denote the opportunistic sender's expected utility when he observes signal s and sends a message that induces public belief β . The key property is:

$$\text{for } s = 1, \quad U(1, \beta) \text{ is decreasing in } \beta.$$

A sender with a good signal wants the public to be pessimistic because he wants to buy cheaply. Conversely, a sender with a bad signal wants the public to be optimistic.

The model also imposes symmetry:

$$U(1, \beta) = U(-1, 1 - \beta).$$

A sender with a good signal facing a pessimistic public is in the same position as a sender with a bad signal facing a correspondingly optimistic public.

These two properties deliver the core strategic-communication structure: the sender's preferred induced belief is opposite to the signal he observes.

5.3 Pre-announcement speculation

The paper also distinguishes another scheme. In pre-announcement speculation, the sender trades before making the announcement and then uses the announcement to move the market price in the desired direction. This scheme does not require private information about the asset payoff; it relies on the sender's ability to move prices.

For example, the sender could first buy the asset, then issue a positive forecast, and profit from the price increase. Alternatively, he could short the asset and issue a negative forecast.

5.4 Why the paper focuses on post-announcement speculation

Post-announcement speculation is more directly connected to insider information. It distorts the information content of prices because the sender uses a private signal to decide which false message is most profitable. It is also often more profitable per share than pre-announcement manipulation.

The benchmark model therefore assumes:

$$\text{the sender trades only after his announcement.}$$

The extension section later shows that the main results survive when both pre- and post-announcement speculation are allowed, provided that there are enough noise traders to make pre-announcement speculation risky.

6 Single-period equilibrium

6.1 Definition of equilibrium credibility

In the one-period game, the sender chooses a message as a Stackelberg leader, but the public anticipates the opportunistic sender's strategy. Equilibrium credibility must be self-fulfilling:

$$\pi = \rho p + (1 - \rho) \Pr(\text{opportunistic sender's message is correct}).$$

Because the opportunistic sender's payoff is decreasing in the induced belief when the signal is good, he wants to lie if his message is believed.

6.2 Static equilibrium result

The paper's first proposition gives a sharp static result.

- If $\rho \leq 1/2$, messages are not credible and no information is transmitted:

$$\pi = \frac{1}{2}.$$

The public ignores the forecast.

- If $\rho > 1/2$, the message is credible even though the opportunistic type always lies:

$$q = 0, \quad \pi = \rho p + (1 - \rho)(1 - p) > \frac{1}{2}.$$

6.3 Intuition

The logic is subtle but simple. If the public thinks the message is informative, an opportunistic sender wants to move the price in the wrong direction and therefore lies. Rational traders understand this. As a result, the only reason the message remains credible is that the sender may be honest. This requires $\rho > 1/2$. If there is less than a fifty percent chance that the sender is honest, the public cannot rationally follow the recommendation.

The static model therefore says that a low-reputation sender cannot move the market. The dynamic model changes this conclusion because reputation can be rebuilt and because the opportunistic type may sometimes tell the truth.

7 Repeated reputation game

7.1 Timing and state variables

In the repeated game, the same sender operates every period. At the beginning of the game, the public believes the sender is honest with probability ρ_1 . Each period:

1. the sender observes a private signal $\tilde{s}$,
2. he sends a public message $\tilde{m}$,
3. market participants update beliefs and prices,
4. the sender trades,
5. the true asset payoff state $\tilde{n}$ is revealed,

6. the public updates the sender's reputation.

The public's belief at the beginning of period t , denoted ρ_t , is the state variable of the dynamic game.

7.2 Credibility in the dynamic game

At reputation ρ_t , if the opportunistic sender tells the truth with probability $q(\rho_t)$, credibility is

$$\pi(\rho_t) = \rho_t p + (1 - \rho_t) [pq(\rho_t) + (1 - p)(1 - q(\rho_t))].$$

This is the probability that the forecast is correct from the public's perspective.

7.3 Bayesian reputation updating

After a correct forecast, the posterior probability that the sender is honest rises to

$$\rho_{t+1}^+ = \frac{\rho_t p}{\pi(\rho_t)}.$$

After an incorrect forecast, reputation falls to

$$\rho_{t+1}^- = \frac{\rho_t(1 - p)}{1 - \pi(\rho_t)}.$$

These formulas summarize why noisy signals create gradual and reversible reputation dynamics. Even an honest sender can be wrong with probability $1 - p$, so an incorrect forecast does not reduce reputation to zero. Even an opportunistic sender can be right, so a correct forecast does not prove honesty.

From the public's subjective perspective, reputation is a martingale:

$$\mathbb{E}[\rho_{t+1} \mid \mathcal{F}_t] = \rho_t.$$

The public's belief is correctly updated given its information.

7.4 Truthful versus lying value

Let $W(\rho)$ be the opportunistic type's continuation value at reputation ρ . If he observes $\tilde{s} = 1$ and tells the truth, he receives

$$V_T(\rho) = U(1, \pi(\rho)) + \delta [pW(\rho^+) + (1 - p)W(\rho^-)].$$

If he lies, he receives

$$V_L(\rho) = U(1, 1 - \pi(\rho)) + \delta [pW(\rho^-) + (1 - p)W(\rho^+)].$$

The difference between these values captures the trade-off:

$$V_L(\rho) - V_T(\rho) = \underbrace{U(1, 1 - \pi) - U(1, \pi)}_{\text{current manipulation gain}} - \underbrace{\delta(2p - 1)[W(\rho^+) - W(\rho^-)]}_{\text{reputation cost of lying}}.$$

The opportunistic type lies when the short-run trading gain exceeds the discounted reputational cost.

7.5 Equilibrium characterization

An equilibrium consists of a credibility function $\pi(\rho)$, a reporting rule $q(\rho)$, and a value function $W(\rho)$ such that:

- If $V_T(\rho) > V_L(\rho)$, the opportunistic type reports truthfully at that reputation.
- If $V_T(\rho) < V_L(\rho)$, he lies.
- If $V_T(\rho) = V_L(\rho)$, he randomizes.
- $W(\rho) = \max\{V_T(\rho), V_L(\rho)\}$.
- The public's credibility belief $\pi(\rho)$ is consistent with the reporting rule and Bayes' rule.

The paper focuses on symmetric Markov equilibria with continuous nondecreasing $W(\rho)$.

7.6 Main dynamic equilibrium theorem

The paper proves that there is a unique symmetric Markov equilibrium with a continuous nondecreasing value function. In that equilibrium:

- The opportunistic type always lies with positive probability.
- Messages always have some credibility for any positive reputation:

$$\pi(\rho) > \frac{1}{2} \quad \text{for all } \rho > 0.$$

- Therefore, information is transmitted and prices respond even when reputation is low.

This result differs sharply from the one-period game. In the static game, a sender with reputation below one-half cannot move the market. In the dynamic game, the sender can still move the market because the possibility of future reputation building makes some truth-telling incentive present.

8 Equilibrium strategies and credibility

8.1 Low reputation

When $\rho = 0$, the public is certain the sender is opportunistic. The paper shows:

$$q(0) = \frac{1}{2}, \quad \pi(0) = \frac{1}{2}.$$

At exactly zero reputation, messages are uninformative. The sender cannot move prices through announcements and only trades on inside information.

For small but positive reputation, however, the sender's message has some credibility. A long sequence of correct forecasts can rebuild reputation. This prevents credibility from collapsing immediately in the dynamic game.

8.2 High reputation

At high reputation levels, the opportunistic type always lies. There exists a threshold $\bar{\rho} < 1$ such that for $\rho > \bar{\rho}$:

$$q(\rho) = 0.$$

The intuition is that a high reputation is valuable because it lets the sender move prices strongly. But near $\rho = 1$, an incorrect prediction has little effect on reputation: if the public is almost certain the sender is honest, a bad forecast is easily interpreted as an honest signal error. Hence the reputational cost of lying becomes small. The sender milks his reputation.

8.3 Intermediate reputation

For intermediate reputation, the opportunistic sender may mix. He sometimes tells the truth to preserve or rebuild credibility and sometimes lies to exploit it. The exact shape of $q(\rho)$ and $\pi(\rho)$ depends on the curvature of the value function and the sender's payoffs. The paper does not impose a simple monotonic rule for all intermediate values.

8.4 Credibility and truthfulness are different

A key conceptual point is that credibility and truthfulness need not move together. Credibility is the probability that a forecast is correct, while truthfulness is the probability that the opportunistic type reports his signal honestly. A higher reputation can raise credibility even if it lowers the opportunistic type's truthfulness, because the message is more likely to come from an honest type.

9 Long-run reputation dynamics

9.1 Fixed type

First suppose the sender's type is fixed forever. If he is actually honest, his forecasts are correct with probability p . Because the public's expected forecast accuracy $\pi(\rho_t)$ is below or equal to p , an honest sender's reputation tends to rise. More formally, ρ_t is a submartingale under the honest type.

If the sender is actually opportunistic, he lies with positive probability. Therefore, his forecasts are correct less often than the public would expect from an honest sender. His reputation tends to decline, and ρ_t is a supermartingale under the opportunistic type.

The main theorem is:

$$\rho_t \rightarrow 1 \quad \text{almost surely if the sender is honest,}$$

$$\rho_t \rightarrow 0 \quad \text{almost surely if the sender is opportunistic.}$$

9.2 Interpretation

In the very long run, the public learns the sender's true type. The opportunistic sender's credibility is a capital stock that he gradually spends down. Eventually, prices stop responding to his messages and he is reduced to ordinary silent insider trading.

However, the result is not a complete defense of reputation as regulation. The learning process can be slow. During the learning phase, the opportunistic type repeatedly distorts prices and rational traders are repeatedly deceived.

9.3 Changing type

The more realistic extension allows the sender's type to change over time. For example:

- different journalists may occupy the same platform in different periods,
- a manager's ability to trade secretly may vary,
- a monitoring process may replace some dishonest agents but not perfectly,
- incentives may change over time.

Let the type $J_t \in \{H, O\}$ follow a Markov process:

$$\Pr(J_{t+1} = H \mid J_t = H) = \phi,$$

$$\Pr(J_{t+1} = O \mid J_t = O) = \psi,$$

with persistence $\phi + \psi > 1$. The public first updates reputation based on the forecast outcome and then adjusts for the possibility that the type changed.

9.4 Long-run implication with changing types

With type renewal, the sender's current type is never fully known. Reputation does not converge to zero or one. Instead, the joint type-reputation process converges to an invariant distribution. In this environment:

- credibility does not vanish,
- information continues to be transmitted,
- opportunistic types continue to have opportunities for manipulation,
- market manipulation can persist indefinitely.

This is the paper's strongest practical warning. Even if individual reputations eventually reveal fixed types, institutions with turnover or changing incentives may never fully discipline manipulation through learning alone.

10 Extensions

10.1 State-dependent payoffs

The paper extends the model so that the sender's payoff can depend on an exogenous state variable θ_t , publicly observed at the beginning of the period. If a higher θ_t increases the current gain from lying, then the equilibrium truth-telling probability and credibility decline with θ_t .

One interpretation is sender influence. If more rational speculators listen to the sender, his message has a bigger price impact, raising the return to manipulation. Under logarithmic preferences and suitable assumptions, the paper shows that greater influence reduces the opportunistic type's truthfulness and the credibility of his announcements.

10.2 Penalties for lying

Suppose a sender who lies faces an expected penalty K . A larger penalty reduces the payoff from dishonesty. The model implies that truthfulness and credibility increase with K . This is intuitive, but the paper emphasizes a limitation: proving that a sender lied is hard when private signals are noisy.

If a forecast is wrong, the sender can claim that his signal was simply incorrect. Therefore, legal penalties for lying may be hard to enforce. This motivates trading restrictions and disclosure requirements, which are easier to verify than truthfulness.

10.3 Combining pre- and post-announcement manipulation

The benchmark model assumes trading only after the announcement. The extension allows both pre-announcement and post-announcement speculation.

If the sender observes a positive signal, he may first buy before the announcement, then announce good news to raise the price, and later adjust his position using his private information. The problem is more complex because pre-announcement speculation depends on the sender's ability to move prices, while post-announcement speculation depends on the difference between the induced belief and the true signal.

The paper shows that, under logarithmic utilities and enough noise traders, all previous results remain valid. Noise traders are crucial because they make pre-announcement manipulation risky. Without enough noise, the sender could manipulate almost risklessly.

11 Figures and model interpretation

11.1 Figure I and Figure II: Timing and manipulation schemes

Read:

- Figure I lays out the period timing. The sender first observes a private signal. He then sends a public message that induces a belief among rational traders. A noise-trader shock is realized at the same time. The asset payoff is revealed only at the end of the period.
- The price before the announcement is uninformative, $P_0 = 1/2$. The price after the announcement is $P(\beta)$, where β reflects the public's posterior belief after the message and noise shock.
- Figure II illustrates the two manipulation schemes for a good private signal $\tilde{s} = +1$. If the sender reports truthfully, the market price rises; if he lies, the price falls. The sender's post-announcement trading return is highest when he first depresses the price and then buys.
- The figure also separates pre-announcement and post-announcement speculation. Pre-announcement gains depend on moving prices through credibility. Post-announcement gains depend on using private information after distorting the public belief.

Economic message: The figures make the paper's mechanism transparent. The sender's private signal lets him know the direction in which the asset is likely to move, while his credibility lets him temporarily move the price in the opposite direction. This creates the incentive to lie: public credibility makes manipulation profitable.

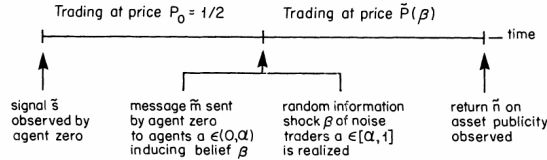


FIGURE I
Timing of Moves and Information

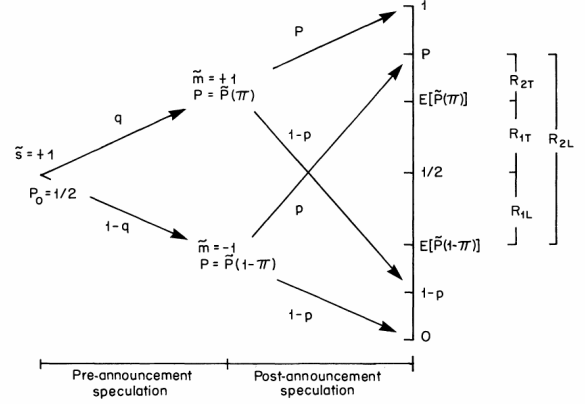


FIGURE II
The Two Market Manipulation Schemes (for $\tilde{s} = +1$)
 R_i is agent zero's expected return (as of the beginning of the period) from speculation in subperiod $i = 1, 2$, when he decides to report truthfully (R_{iT}) or to lie (R_{iL}).

- (a) Figure I: timing of moves and information. (b) Figure II: the two market manipulation schemes.

12 Policy implications

12.1 Why reputation is not enough

The paper does not claim that reputation is irrelevant. Reputation does discipline the sender. An opportunistic type cannot lie every period without eventually damaging credibility. However, reputation is incomplete discipline for three reasons.

- **Noisy signals create plausible deniability.** A bad forecast is not proof of a lie.
- **Manipulation can be profitable before discovery.** Even if reputation eventually falls, the sender can earn trading profits during the learning phase.
- **Types and incentives change.** If the identity or incentives of the informed agent change over time, reputation may never fully reveal the current type.

12.2 Why trading restrictions may be useful

The model gives a rationale for restrictions on trading by market-moving insiders. If a manager, journalist, or guru is allowed to trade the asset affected by his announcements, he has an incentive to distort public information. Since truthfulness is hard to verify, restrictions on trading or prompt disclosure of trades may be more effective than after-the-fact fraud enforcement.

The central logic is:

noisy signal $\Rightarrow$ hard to prove lying $\Rightarrow$ reputation insufficient $\Rightarrow$ need trading constraints.

12.3 What kind of policy?

The paper suggests several possible interventions:

- require insiders who frequently issue forecasts to promptly disclose trades,
- limit trading in assets directly affected by those forecasts,

- impose penalties for misrepresentation when lying can be established,
- reduce the ability to trade secretly through monitoring and audit rules.

The paper emphasizes that mandatory disclosure may still be imperfect because of disclosure delays and because insiders may claim liquidity motives. In some cases, outright trading restrictions may be needed.

13 How the paper relates to earlier literature

13.1 Relation to insider-trading models

Traditional insider-trading models such as Grossman–Stiglitz and Kyle focus on information entering prices through trading. In those models, an insider trades on private information, and prices partially reveal the information through order flow. Benabou and Laroque focus on a different channel: the insider publicly announces information, possibly false, and the announcement itself moves prices.

This difference matters because the sender’s communication strategy becomes central. The insider is not just hiding information; he is actively manipulating the public signal.

13.2 Relation to strategic communication

The model builds on Crawford–Sobel and Sobel-style strategic communication. The sender observes information, sends a message, and receivers choose an action. Here the receivers’ action is embodied in market price. The sender’s preferences are opposed to receivers’ information needs because he profits from moving the price away from the direction suggested by his signal.

The paper’s technical contribution is to allow noisy private signals. This modification makes reputation dynamics more realistic because incorrect forecasts do not perfectly reveal dishonesty.

13.3 Relation to reputation models

Classic reputation models often feature a type that is revealed after a clear deviation. In this paper, the sender’s reputation changes gradually. Because signals are noisy, a single bad prediction reduces reputation but does not destroy it. This creates a richer and more realistic reputation process with both upward and downward movements.

14 Short summary

- The paper studies an informed sender whose public statements move asset prices and who can trade after making those statements.
- The sender observes a noisy signal about an asset payoff. Because the signal is noisy, incorrect predictions can be honest mistakes.
- The public gives the sender credibility because he may be honest and because opportunistic types sometimes tell the truth to preserve reputation.
- In a one-period game, messages matter only if the prior probability of honesty exceeds one-half. In that case, the opportunistic type always lies, but credibility remains positive because the sender may be honest.

- In the infinite-horizon game, the opportunistic type always lies with positive probability, but he may sometimes tell the truth to protect reputation.
- Reputation updates upward after correct forecasts and downward after incorrect forecasts, but never collapses immediately because signals are imperfect.
- If the sender's type is fixed, the public eventually learns whether he is honest or opportunistic.
- If the sender's type changes over time, learning remains incomplete and manipulation persists indefinitely.
- The policy implication is that trading restrictions and trade disclosure rules can be necessary when truthfulness is hard to verify.

15 Conclusion

15.1 Core conclusion

The paper shows that privileged information can generate not only profitable insider trading but also market manipulation through public announcements. An informed sender who can move prices and then trade has incentives to announce the opposite of his signal. Because private information is noisy, wrong forecasts do not conclusively reveal lies, and reputation disciplines manipulation only gradually.

15.2 Economic intuition

The sender's credibility turns public announcements into a pricing technology. If he is believed, he can move prices. If he also has private information, he knows whether the price has been moved in a profitable direction. The most profitable use of credibility is often to distort the price away from fundamentals and then trade against the distortion.

15.3 Incremental contribution revisited

The paper's main contribution is to integrate three elements: noisy private information, strategic market-moving announcements, and dynamic reputation. This combination explains why opportunistic insiders can manipulate repeatedly without being fully exposed after a single bad forecast.

15.4 Implications for market design

The model supports the idea that information integrity in asset markets may require rules beyond ordinary anti-fraud enforcement. Since a lie is hard to prove when forecasts are noisy, regulators may need to focus on verifiable behavior: trades, disclosure timing, conflicts of interest, and restrictions on the ability of forecasters to trade assets affected by their statements.

15.5 Limitations

The model is deliberately stylized. It uses two payoff states, a short-term asset, a pure speculative market, and a simplified division between rational and noise traders. The authors acknowledge that richer settings with positive asset supply, multiple states, persistent securities, and real investment effects remain open questions. Still, the basic mechanism is robust: when public messages

move prices and truthfulness is imperfectly verifiable, private information can become a tool for manipulation.

15.6 Takeaway

Credibility does not eliminate manipulation; it can enable it. Reputation is valuable because it makes announcements move prices, and that is exactly why an opportunistic insider may want to spend credibility by lying. When signals are noisy and types change, market manipulation can survive even in the long run.